

Image Processing – Fourier Extraction

Finger-Vein PAD using Fourier Magnitude and Phase Features

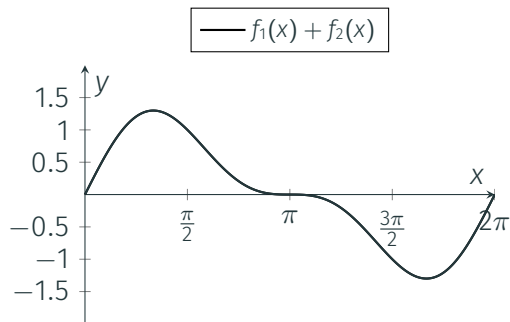
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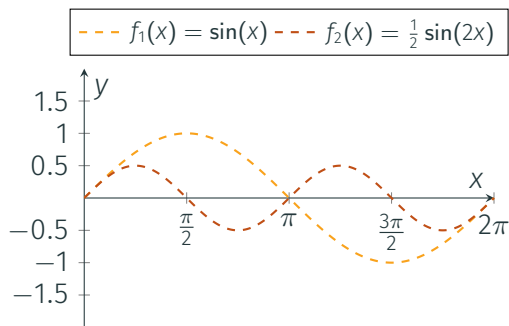
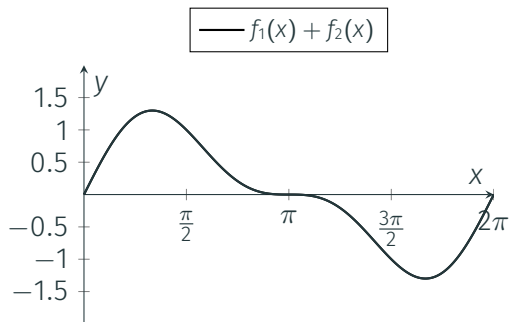
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- Task 1
 - Experiments
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- Task 2
 - Experiments
 - Results
- Final Results & Conclusion

Fourier Introduction

GRAPHICAL REPRESENTATION OF FOURIER



GRAPHICAL REPRESENTATION OF FOURIER



Fourier-Series (approx. formula)

$$f(t) \approx \sum_{k=-\infty}^{\infty} c_k e^{i\omega_k t}$$

- k : index of the frequency components
- ω_k : circular frequency of the k -th vibration
- $c_k \in \mathbb{C}$: complex fourier coefficients

$$c_k = \operatorname{Re}(c_k) + i \operatorname{Im}(c_k)$$

$$|c_k| = \sqrt{\operatorname{Re}(c_k)^2 + \operatorname{Im}(c_k)^2} \quad (\text{Magnitude})$$

$$\varphi_k = \arctan \left(\frac{\operatorname{Im}(c_k)}{\operatorname{Re}(c_k)} \right) \quad (\text{Phase})$$

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Magnitude $|c_k|$

→ Strength of the frequency ω_k .

Phase φ_k

→ Gives a “position” relative to the x-Axis.

WHAT ARE WE DOING?

- **Fourier** feature extraction.
→ 2D Fast-Fourier-Transformation (**FFT**)